MAHATMA GANDHI UNIVERSITY, KOTTAYAM
MGU-UGP (Honours)/ MGU-BBA (Honours)/ MGU-BCA (Honours)
REGULAR EXAMINATION October 2026

Fifth SEMESTER
Core Course (CC) - MG5CCRBBA301 - LOGISTICS AND SUPPLY CHAIN MANAGEMENT
(2024 ADMISSION ONWARDS)

Duration: 1hour 30 minutes Maximum Marks: 50

Remember(K), Understand(U), Apply(A), Analyse(An), Evaluate(E), Create(C), Skill(S), Interest(I) and Appreciation(Ap)

Students should attempt at least one question from each course outcome to enhance their overall outcome attainability.

Part A
Short Answer Questions(four to five sentences)
A choice between two options for each question, answer five short answer questions, one from each pair

1a. How do facilities and transportation influence supply chain network decisions? [U] / [CO1]

1. **Facilities:** The number, location and capacity of factories, warehouses and distribution centres determine where products are manufactured, stored and supplied to customers. Proper facility locations help reduce costs and improve customer service.
2. **Transportation:** Transportation connects suppliers, factories, warehouses and customers. The choice of transport mode, such as road, rail, air or sea, affects delivery time, transportation costs and the movement of goods.
3. **Overall impact:** Companies must balance facility costs and transportation costs to build an efficient supply chain. For example, opening more warehouses may reduce delivery time but increase storage costs.

OR

1b. A retailer experiences a small increase in customer demand, but the manufacturer receives a much larger increase in orders. Identify the supply chain problem and state its meaning. [U] / [CO1]

The **bullwhip effect** occurs when a small change in customer demand leads to a much larger change in orders as they move up the supply chain from retailers to manufacturers and suppliers.

For example, if a retailer notices a slight increase in product sales and orders extra stock to avoid shortages, the manufacturer may interpret this as a major increase in demand and produce more than necessary. This can lead to excess inventory, higher costs and production problems.

2a. Distinguish qualitative and quantitative forecasting and identify one suitable situation for each. [U] / [CO2]

1. Qualitative forecasting: It is based on human judgment, experience and expert opinions rather than mainly on numerical data. Example: Forecasting demand for a new product when past sales data are not available.
2. Quantitative forecasting: It uses past sales data, numerical information and statistical methods to estimate future demand. Example: Using previous months' sales figures to forecast demand for a product during the next month.

OR

2b. Explain the contribution of capacity utilisation to revenue management. [U] / [CO2]

Capacity utilisation helps revenue management by ensuring that available resources, such as factory capacity, warehouse space and transport vehicles, are used effectively to generate more revenue. When utilisation is low, a company can offer discounts to attract customers and make better use of unused capacity. During peak demand, it can adjust prices and allocate capacity to higher-paying customers to increase profits. Thus, proper capacity utilisation helps balance demand and supply while improving profitability.

3a. A business wants its marketing, procurement, production, and logistics departments to work together efficiently. Explain how integrating supply activities across business functions can help achieve this objective [U] / [CO3]

Integrating supply activities means coordinating the work of marketing, procurement, production, and logistics departments rather than allowing each department to work separately.
Marketing shares customer demand information with procurement and production so that materials can be purchased and goods produced in the required quantities.
Logistics coordinates the storage and delivery of materials and finished products. This teamwork reduces delays, avoids excess inventory, and controls operating costs. As a result, the business can meet customer requirements more efficiently.

OR

3b. Discuss inventory control systems and their importance. [U] / [CO3]

Inventory control systems are methods used by a business to monitor stock levels and decide when and how much to order.
Common systems include the ABC analysis, Economic Order Quantity (EOQ), and the reorder point system.
These systems help maintain sufficient stock to meet customer demand without keeping too much inventory. They also reduce storage costs, prevent stock shortages, and avoid unnecessary investment in stock. Thus, inventory control helps a business use its resources efficiently and ensure the smooth flow of goods.

4a. Explain how transportation costs influence supply chain network design. [U] / [CO1]

Transportation costs play an important role in deciding the location of factories, warehouses, and distribution centres in a supply chain. When transportation costs are high, businesses may establish warehouses closer to customers to reduce delivery distances and expenses. When transportation costs are low, they may prefer fewer, centrally located facilities to reduce warehouse and operating costs. Therefore, businesses must balance

transportation costs with inventory, warehousing, and customer service costs when designing their supply chain network.

OR

4b. A company is facing high transportation costs because its warehouses are located far from its customers. Explain how network design can help the company . [U] / [CO1]

Network design helps the company review the location of its warehouses and identify better locations closer to customers. By opening warehouses or distribution centres in suitable locations, the company can reduce delivery distances, fuel expenses, and transportation time. It can also use customer demand and transportation cost information to decide the appropriate number and location of warehouses. However, the company should balance transportation savings against additional warehouse and inventory costs. This helps reduce total supply chain costs while improving customer service.
For example, if a company has one warehouse in Chennai supplying customers across Tamil Nadu, it could open an additional distribution centre near Madurai to serve customers in southern districts more quickly.

5a. A hotel regularly purchases stationery, cleaning chemicals and replacement bulbs. Classify these purchases and explain how their sourcing priorities may differ. [An] / [CO2]

The hotel's purchases can be classified based on their cost, importance to daily operations, and availability from suppliers.
Stationery is generally a routine item, so the hotel should focus on simple purchasing procedures and low administrative costs.
Cleaning chemicals are regular consumables, and the hotel should compare prices while ensuring quality and a reliable supply to maintain hygiene.
Replacement bulbs are also routine items when standard bulbs are readily available, but special bulbs needed for essential areas may require extra safety stock.
Thus, the hotel should give greater attention to cost, quality, and supply risk according to the nature of each purchase.

OR

5b. Identify criteria commonly used in supplier selection. [U] / [CO2]

Supplier selection is based on criteria such as price, quality, delivery reliability, production capacity, and service. For example, a supplier offering a lower price may not be the best choice if frequent delivery delays interrupt production. Therefore, analysing these criteria helps a business select suppliers who offer the best overall value and reliability.

[5x3 = 15]

Part B
Long Essay
A choice between two options for each question, answer two short essay questions, one from each pair

6a. A manufacturing company wants to improve the movement, storage and delivery of its products. Explain how Logistics Management can help the company and discuss its importance. [U] / [CO1]

Logistics management refers to the planning, implementation, and control of the movement and storage of goods from the point of origin to the point of consumption. In a manufacturing company, it helps ensure that raw materials reach the factory on time and finished products are delivered to customers safely and without unnecessary delays. Its major activities include transportation, warehousing, inventory management, material handling, and order processing. – ***3 marks***

The company can improve its operations in the following ways: ***- 5 marks***

1. Transportation management: Selecting suitable transport modes and planning efficient delivery routes helps reduce transportation costs and delays.
2. Warehousing: Proper warehouse planning provides safe storage for raw materials and finished products and makes goods easier to locate and handle.
3. Inventory management: Maintaining appropriate stock levels prevents shortages of raw materials and avoids excess inventory and unnecessary storage costs.
4. Material handling: Using suitable equipment and handling methods helps move goods safely within the factory and warehouse, reducing damage and wastage.
5. Order processing: Recording customer orders accurately and processing them quickly helps ensure that the correct products are delivered on time.
6. Cost reduction: Coordinating transportation, storage, and inventory activities helps control overall logistics costs and use company resources efficiently.
7. Customer satisfaction: Timely delivery of the right products in good condition improves customer confidence and encourages repeat purchases.

Thus, logistics management helps a manufacturing company maintain a smooth flow of materials and finished goods, reduce operating costs, and provide better customer service

Examples and conclusion – 2 marks

6b. A manufacturing company currently uses only road transportation to move raw materials and finished goods. Increasing fuel costs and delivery delays have affected its supply chain performance. Apply your knowledge of transportation modes and transportation choice to examine whether the company should consider other modes. Explain how cost, speed, distance, reliability and nature of goods can influence its decision. [A] / [CO1]

The company should consider using other transportation modes because rising fuel costs and delivery delays are affecting its performance. Instead of depending entirely on road transport, it can combine road with rail or air, depending on the type of goods and delivery requirements. Rail transport may reduce the cost of moving heavy raw materials over long distances, while road transport can be retained for short-distance deliveries and final distribution. Air transport may be suitable for urgent, high-value components, although its cost is much higher. ***- 2 marks***

The company should apply the following factors when selecting the most suitable transportation mode. – ***with explanation 5 marks***

1. Cost: The company should compare freight charges, fuel expenses, handling costs, and other related expenses. For example, it can transport bulk raw materials by rail to reduce long-distance transport costs and use trucks for the final delivery to its factory.

2. Speed: If customers need finished products urgently, the company should choose a faster mode where the additional cost is justified. Air transport may suit urgent, high-value components, while rail may be more suitable for less urgent bulk shipments.

3. Distance: Road transport is convenient for short distances because it provides door-to-door service. For long-distance movement of heavy materials, the company should examine rail transport, provided suitable rail terminals and connections are available.

4. Reliability: Since road traffic is causing delays, the company should compare the delivery schedules and consistency of alternative transport services. It can use scheduled rail services for regular bulk shipments while retaining trucks for flexible deliveries and unexpected requirements.

5. Nature of goods: Heavy materials such as steel and cement are generally suitable for rail transport, whereas fragile, high-value components needed quickly may justify air transport. Perishable or hazardous goods require suitable handling facilities and safety arrangements.

Example: A manufacturing company purchasing steel from another state can transport large quantities by rail to a terminal near its factory and use trucks for the final delivery. It can continue using road transport for local suppliers and urgent orders. This combination may reduce transportation costs without sacrificing delivery flexibility. – ***3 marks***

7a. Explain the importance of supply chain drivers and metrics. Discuss how appropriate performance measures can help an organization understand and improve its supply chain.

Importance of Supply Chain Drivers and Metrics- 2 marks

Supply chain drivers are the factors that influence how efficiently and quickly a product moves from suppliers to customers. The main drivers are facilities, inventory, transportation, information, sourcing, and pricing. Supply chain metrics are measures used to check how well these activities are performing. They help an organisation identify problems, control costs, improve delivery performance, and meet customer needs.

Major supply chain drivers and their performance measures- Explains any four relevant drivers with suitable measures ***(1 mark each)***

1. Facilities 2. Inventory 3. Transportation 4. Information 5. Sourcing 6. Pricing

How performance metrics help improve the supply chain -*Explains identification of problems, cost control, customer service, or better decision-making* ***(any two points, 1 mark each)***

- Identify problems; Control costs; Improve customer service; Support better decisions; Measure progress

Example Gives a relevant example showing how metrics improve performance ***1 marks***

Conclusion Summarises how drivers and metrics support supply chain efficiency ***1 marks***

7b. A company has sufficient warehouse space but still experiences frequent stock shortages and delays in order fulfillment. Analyze the possible reasons and explain how warehouse design, warehouse decisions, and logistics management can improve the situation. [U] / [CO1]

Answer	content	Marks
Introduction and possible reasons	Explains why shortages and fulfillment delays occur despite sufficient warehouse space	2
Warehouse design	Layout, product placement, picking, packing, and dispatch areas	2
Inventory control and warehouse decisions	Stock accuracy, reorder levels, safety stock, and storage decisions	2
Logistics management	Coordination of suppliers, replenishment, warehousing, and transportation	2

Example and conclusion	Relevant example and summary of expected improvements	2
Total		10

[2x10 = 20]

Part C
Compulsory Case Study

8. Kerala Mart purchases products from local suppliers and distributes them to stores across Kerala. Recently, the company has faced demand fluctuations, supplier delays, transportation problems, rising costs, and increasing customer expectations. Different departments are also making decisions independently. Answer the following: a) How can KeralaMart apply integrated supply chain planning to coordinate its activities? b) Explain how forecasting, sourcing, inventory, and distribution decisions can be synchronized. c) How can cross-functional integration and supply chain responsiveness help KeralaMart manage its current [A] / [CO3]

[1 x 15 = 15]

Question	Answer	Marks
a) Integrated supply chain planning	• Meaning and need for integration • Sharing information and coordinating departments • Suitable application to KeralaMart	5
b) Synchronising supply chain decisions	• Forecasting based on demand patterns • Sourcing and supplier coordination • Inventory and safety stock planning • Distribution and delivery scheduling • Suitable example showing synchronisation	5
c) Cross-functional integration and responsiveness	• Explains cross-functional integration and its benefits. • Explains responsiveness and practical measures. • Suitable application to KeralaMart's challenges.	5